

Experiment-19:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <ctype.h>
```

```
char productions[20][20];
```

```
char leading[20][20];
```

```
char nonTerminals[20];
```

```
int n;
```

```
int findIndex(char c)
```

```
{
```

```
    for(int i = 0; i < n; i++)
```

```
    {
```

```
        if(nonTerminals[i] == c)
```

```
            return i;
```

```
    }
```

```
    return -1;
```

```
}
```

```
void addLeading(int index, char c)
```

```
{
```

```
    if(strchr(leading[index], c) == NULL)
```

```
    {
```

```
        int len = strlen(leading[index]);
```

```
        leading[index][len] = c;
```

```
        leading[index][len + 1] = '\0';
```

```
    }
```

```
}
```

```

void computeLeading(char c)
{
    int index = findIndex(c);

    for(int i = 0; i < n; i++)
    {
        if(productions[i][0] == c)
        {
            char symbol = productions[i][3];

            // If first symbol is terminal
            if(!isupper(symbol))
            {
                addLeading(index, symbol);
            }

            // If first symbol is non-terminal
            else
            {
                int next = findIndex(symbol);

                // Copy already calculated leading symbols
                for(int j = 0; leading[next][j] != '\0'; j++)
                {
                    addLeading(index, leading[next][j]);
                }

                // Check next terminal after non-terminal
                if(productions[i][4] != '\0' &&
                    !isupper(productions[i][4]))
                {

```

```

        addLeading(index, productions[i][4]);
    }
}
}
}
}

```

```

int main()
{
    printf("Enter number of productions: ");
    scanf("%d",&n);

    printf("Enter productions:\n");

    for(int i=0;i<n;i++)
    {
        scanf("%s",productions[i]);
        nonTerminals[i]=productions[i][0];
        leading[i][0]='\0';
    }

    // Repeat calculation to update sets
    for(int k=0;k<n;k++)
    {
        for(int i=0;i<n;i++)
        {
            computeLeading(nonTerminals[i]);
        }
    }

    printf("\nLEADING Sets:\n");

```

```

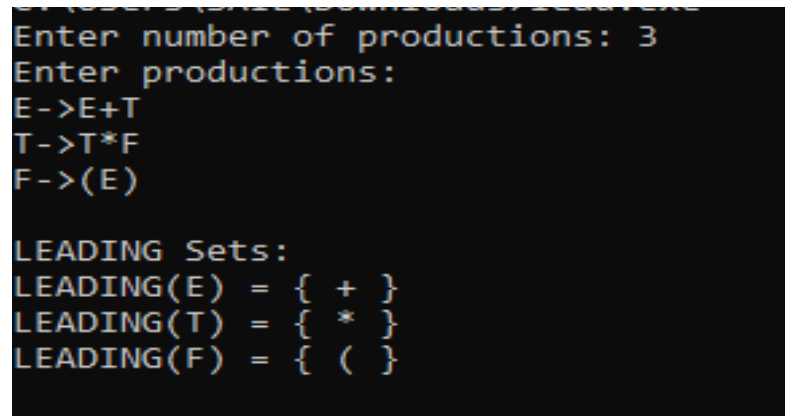
for(int i=0;i<n;i++)
{
    printf("LEADING(%c) = { ",nonTerminals[i]);

    for(int j=0;leading[i][j]!='\0';j++)
    {
        printf("%c ",leading[i][j]);
    }

    printf("}\n");
}

return 0;
}

```



```

Enter number of productions: 3
Enter productions:
E->E+T
T->T*F
F->(E)

LEADING Sets:
LEADING(E) = { + }
LEADING(T) = { * }
LEADING(F) = { ( }

```